

Machine Learning-Based Risk Analysis for Construction Worker Safety from Ubiquitous Site Photos and Videos

Shuai Tang, A.M.ASCE¹; and Mani Golparvar-Fard, A.M.ASCE²

Abstract: This paper proposes a new method for single-worker severity level prediction from already collected site images and video clips. Onsite safety observers often assess workers' severity levels during construction activities. While risk analysis is key to improving long-term construction site safety, omnipresent monitoring is still time-consuming and costly to implement. The recent growth of visual data captured actively on construction sites has opened a new opportunity to increase the frequency of worker safety monitoring. This paper shows that a comprehensive vision-based assessment is the most informative to automatically infer worker severity level from images. Efficient computer vision models are presented to conduct this risk analysis. The method is validated on a challenging image dataset first of its kind. Specifically, the proposed method detects and evaluates the worker state from visual data, defined by (1) worker body posture, (2) the usage of personal protective equipment, (3) worker interactions with tools and materials, (4) the construction activity being performed, and (5) the presence of surrounding workplace hazards. To estimate the worker state, a multitasked recognition model is introduced that recognizes objects, activity, and keypoints from visual data simultaneously, taking 36.6% less time and 40.1% less memory while keeping comparably performances compared to a system running individual models for each subtask. Worker activity recognition is further improved with a spatio-temporal graph neural network model using recognized per-frame worker activity, detected bounding boxes of tools and materials, and estimated worker poses. Finally, severity levels are predicted by a trained classifier on a dataset of images of construction workers accompanied with ground truth severity level annotations. In the test dataset assembled from real-world projects, the severity level prediction model achieves 85.7% cross-validation accuracy in a bricklaying task and 86.6% cross-validation accuracy for a plastering task, demonstrating the potential for near real-time worker safety detection and severity assessment. DOI: [10.1061/\(ASCE\)CP.1943-5487.0000979](https://doi.org/10.1061/(ASCE)CP.1943-5487.0000979). © 2021 American Society of Civil Engineers.

Introduction

The 2019 National Census of Fatal Occupational Injuries report released by the US Bureau of Labor Statistics (BLS) shows that the private construction industry had 1,061 fatal injuries for the year, up 5% from 2018 and the sector's highest number of worker deaths since 2007 (BLS 2020). Exposure to harmful substances or environments caused 167 workers to lose their lives. Another 146 workers lost their lives after making contact with objects or equipment that we deemed unsafe. These statistics are disappointing and make apparent that more work must be done not only to detect and prevent workers from job site hazards but also to improve the total occupational health of the construction workforce.

Having instant access to safety observations helps project managers identify potential safety issues and allows them to proactively monitor their crew engagements. Thanks to the significant advancement made in deploying imaging technologies such as 360-degree, time-lapse, and drone cameras on construction sites (Lin et al. 2015; Han and Golparvar-Fard 2017), an opportunity has merged to deploy computer vision-based techniques at a scale that matters. Building on

the growing availability of these ubiquitous sources of visual data, numerous visual-based safety inspection methods and tools are introduced through academia (Seo et al. 2015; Fang et al. 2020; Zhang et al. 2020a). Startup companies such as Smartvid.io have also deployed similar capabilities in detecting 50 different safety-related objects (e.g., hardhat, glove, glasses, ladder) from images and videos over 1,000 construction sites. A large number of case studies have already been published that show the effectiveness of these solutions across a variety of residential, commercial, and industrial projects. These recent developments fundamentally change the practice of quantitative measurements of incidents and safety benchmarks across many projects at the enterprise level. The wide applicability of these computer vision techniques and the availability of a large volume of visual data has built an opportunity for more advanced and frequent safety assessments, particularly when they are compared with traditional labor-intensive weekly or bi-weekly safety reporting methods. Despite the advantages of these solutions in predicting and preventing incidents, a number of scientific problems have yet to be addressed to fully benefit from the potential of the available visual data and current computer vision techniques:

1. Existing computer vision safety inspection tools explicitly detect safety-related objects and generate their confidence scores, without analyzing (1) worker activities in the context of safety; or (2) assessing incident severity under activities. Existing construction risk models, such as the Construction Hazard Assessment with Spatial and Temporal Exposure (CHASTE) approach (Sacks et al. 2009; Rozenfeld et al. 2010), often describe risk as a function of probability and severity of incidents.
2. Many computer vision safety inspection tools count recognized incidents regardless of the severity levels of different incident occasions. For example, in a plastering task, wet Portland

¹Ph.D. Candidate, Dept. of Civil and Environmental Engineering, Univ. of Illinois at Urbana-Champaign, 205 North Mathews Ave., Urbana, IL 61801 (corresponding author). ORCID: <https://orcid.org/0000-0001-7822-6972>. Email: stang30@illinois.edu

²Associate Professor, Civil Engineering, Computer Science and Technology Entrepreneurship, Univ. of Illinois at Urbana-Champaign, 205 North Mathews Ave., Urbana, IL 61801. Email: mgolpar@illinois.edu

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cement can cause dermal irritation or burns if a worker does not wear proper personal protective equipment (PPE), such as safety glasses and gloves. Applying plaster is more likely to cause harm to naked skin than collecting plaster because of the splatter from applications. Understanding this difference requires comprehensive reasoning on a worker's activity, body pose, and interactions with PPE, tools, and material. However, the detection of safety hazards in this rich context of the workplace is not fully explored. As such, beyond reporting on the fluctuation of safety incidents and their frequency over project timeline, current computer-vision-driven methods are unable to provide insights into the severity of the safety incidents.

To enable risk analysis for safety monitoring, modern computer vision techniques such as human-object interaction (Tang et al. 2020) and object relations (Xiong et al. 2019) are necessary to detect and analyze workers and their interactions with tools, equipment, and the work environment. Also, safety inspection must cover a wider range of tasks including recognizing unsafe activities, unsafe conditions, and occupational health risks (Heinrich et al. 1941; Luttmann et al. 2003; Mitropoulos et al. 2009; Choudhry 2014). However, current computer vision safety inspection methods and tools heavily focus on PPE compliance checking. An underlying technical limitation is that the object detection models introduced in these tools are often incapable of analyzing worker activities to measure the risk of exposure to harmful substances or environments, potential fall-related incidents (Fang et al. 2018c), or unsafe body postures (Yu et al. 2019; Chu et al. 2020). Instead, they primarily focus on pose estimation or activity recognition (Roberts et al. 2020; Luo et al. 2018b; Yan et al. 2019). There is a need for methods that can jointly detect workers, their activities, and their interaction with tools from site images and videos. Such methods allow for modern tools that can minimize the inefficiencies associated with existing isolated safety inspection tools.

The main contribution of this study is a new machine learning-based method to predict worker-level severity using a single-worker's visual data captured at a close distance. Experiment results show that a combined worker state, including information of the workers' body pose, the activities being performed, their PPE use, their interactions with tools and material, and the presence of workplace hazards, is the most informative for this prediction task. This study also contributes to the body of knowledge by constructing a large and highly diverse construction image dataset first of its kind with abundant annotations in worker activity labels, worker body keypoints, and bounding box annotations of protective equipment, tools, and material. In addition to the contributions, this study also presents two technical improvements. (1) a joint-task model to recognize objects, worker activity, and worker body keypoints from a single image. This model is more efficient and achieves comparable performance than using individual models for each subtask. (2) a spatio-temporal graph neural network model to refine frame-wise worker activity recognition by taking in per-frame object, activity, and keypoint predictions. These two improvements foreshadow a unification of worker activity recognition, worker PPE compliance checking, and worker ergonomic risk monitoring tasks from videos of workers taken at a close distance. Finally, single-worker severity levels are predicted by a trained classifier on a dataset of images of construction workers accompanied with ground truth severity level annotations. The severity level prediction model using the full worker state achieves over 85% cross-validation accuracy on the test data.

The following section reviews the most relevant work on risk analysis models, visual-based unsafe action recognition, safety inspection, and ergonomic risk analysis. The third section introduces

the proposed risk analysis model, breakdown of risk formulation, and human study procedure. The fourth section introduces the vision model used for joint activity recognition, object detection, and pose estimation. The spatio-temporal model for refining activity recognition is also presented. The fifth section introduces a brick-laying and plastering activity dataset used for validation. The sixth section presents the details on model specifications and experiment setups. The seventh section analyzes the experiment results. Finally, the findings from this work, the limitations, and open research areas are discussed.

Related Work

Prior research related to this study can be categorized into two buckets: (1) safety risk analysis; and (2) computer vision-driven techniques for worker safety inspection, ergonomic assessment, and activity recognition. As the present study covers a broad range of topics and techniques, we offer a list of key features and their code names from previous studies and this work. This list and Table 1 summarize previous studies' scopes, objectives, and their main differences with this study. The list of key features in this study are:

- DET: Detect the presence of worker or PPE.
- HOI: Recognize worker interactions or relations.
- 2DWP: Estimate worker pose from images.
- 3DWP: Direct reconstruction of 3D worker pose from a single image.
- AR: Recognize construction activities.
- FAR: Provide frame-level activity labels.
- MT: Learn multiple tasks jointly.
- SP: Model spatio-temporal dependencies.

Safety Risk Analysis Models

A number of prior works on construction risk analysis identify a potential area where risk analysis methods can be integrated with safety inspection practices. Hallowell and Gambatese (2010) propose a safety equilibrium equation, measuring (1) the safety risk as to the sum of all risk scores associated with construction activities conducted in a project; and (2) the *safety capacity* of a safety program. Here the safety capacity refers to the ability of safety program elements to mitigate a portion of the common safety risks (Hallowell and Gambatese 2010). For a construction activity such as climbing a ladder and transporting material, it is associated with a set of frequently observed worker injury causes, such as contact with objects, falls, and overexertion. Risk scores of these causes are summed as the total risk score for that construction activity. Individual risks are measured by the likelihood of injury from the historical records and a predetermined severity scale. The CHASTE risk analysis method (Sacks et al. 2009; Rozenfeld et al. 2010) estimates loss-of-control (unwanted, undesired) event risks by the probability of events, expected severity of damage, crew size, and spatio-temporal degree of exposure to hazards during incidents. Real-time risk monitoring (Årnes et al. 2005) estimates the real-time risk as the expected risk of all possible safety states at each timestamp. In practice, the real-time risk formulation can be applied for both online monitoring and offline analysis of time series data. Results aggregated over an event's duration can be used as the risk score for an activity in Hallowell and Gambatese (2010) and the expected severity level in Sacks et al. (2009). Paltrinieri et al. (2019) and Poh et al. (2018) validated machine learning approaches to classify risk indexes and categories. Their findings show the practicality of learned classification models for estimating severity from safety observations.

Table 1. Summary of the reviewed literature

Scope	Previous studies	Objectives	Risk Assmt.	DET	HOI	2DWP	3DWP	AR	FAR	MT	SP
Safety risk analysis models	Hallowell and Gambatese (2010), Rozenfeld et al. (2010), and Sacks et al. (2009)	Evaluate accident and process-level risks from historical data and observer recordings.	○	—	—	—	—	—	—	—	—
	Paltrinieri et al. (2019), Poh et al. (2018), and Årnes et al. (2005)	Learning-based methods to predict risks.	•	—	—	—	—	—	—	—	—
Computer vision-based safety inspection	Park and Brilakis (2012), Shrestha et al. (2015), Park et al. (2015), Fang et al. (2018b), Nath et al. (2020), and Wu et al. (2019)	Recognize personal protective equipment and safety gear noncompliant instances by the presence of objects.	◐	•	—	—	—	—	—	—	—
	Fang et al. (2018a), Xiong et al. (2019), Tang et al. (2020), and Zhang et al. (2020b)	Recognize personal protective equipment and safety gear noncompliant instances by reasoning object relations.	◐	•	•	—	—	—	—	—	—
Computer vision-based ergonomic risk analysis	Ray and Teizer (2012), Liu et al. (2016), and MassirisFernández et al. (2020)	Analyze worker ergonomic risk from a single image, stereo image pairs, or image-depth data.	•	•	—	•	—	—	—	—	—
	Zhang et al. (2018), Yu et al. (2019), and Chu et al. (2020)	Validate vision-based ergonomic assessment with 3D wearable sensor data; use existing ergonomic assessment systems.	•	•	—	•	•	—	—	—	—
Computer vision-based worker activity recognition	Yang et al. (2016), Luo et al. (2018b, a), and Ding et al. (2018)	Recognize worker activity at clip level using a stack of frames.	◐	•	—	—	—	•	—	—	—
	Luo et al. (2019a), Cai et al. (2019), and Roberts et al. (2020)	Recognize worker activity at frame level; incorporate other visual cues.	◐	•	—	—	—	•	•	—	—

Note: DET = detect the presence of worker or PPE; HOI = recognize worker interactions or relations; 2DWP = estimate worker pose from images; 3DWP = direct reconstruction of 3D worker pose from a single image; AR = recognize construction activities; FAR = provide frame-level activity labels; MT = learn multiple tasks jointly; and SP = model spatio-temporal dependencies. A full solid circle (•) means a fully automated process; a half solid circle (◐) marks a user-driven/interactive process with some automation; an empty circle (○) marks a fully user-driven process with no automation. This list of previous studies may be incomplete. Only the most relevant papers are shown.

This paper also takes a learning approach to estimate severity levels. However, the proposed model is applied on a single-worker level and uses visually recognized information referred to as the worker state as its input features. An approximation heuristic is used to handle a much larger state space than that used by Årnes et al. (2005).

Computer Vision-Based Safety Inspection

A large body of prior work focuses on detecting PPE (e.g., hard hat, safety vest) using construction site visual data. Early work on this topic uses low-level image features such as edges (Shrestha et al. 2015) and histograms of oriented gradients (Park et al. 2015). Classifiers such as the support vector machine and k-nearest neighbors are used on image features (Park and Brilakis 2012). More recently, deep learning-based object detectors have been used to detect hard hats (Fang et al. 2018b; Nath et al. 2020; Wu et al. 2019; Xiong et al. 2019; Tang et al. 2020; Zhang et al. 2020b), safety vests (Nath et al. 2020; Tang et al. 2020), harnesses (Fang et al. 2018a, c; Tang et al. 2020), and many other protections such as safety glasses (Tang et al. 2020). Research has also explored human-object interactions to recognize PPE use in various work conditions including indoors and outdoors (Tang et al. 2020; Xiong et al. 2019; Zhang et al. 2020b). These methods only focus on the recognition task and as such do not offer any measurements on safety risk. Different from prior work, this paper only uses visual-based PPE compliance results as a component of the overall risk analysis by integrating that with an assessment of worker activities, body posture, as well as the work context.

Computer Vision-Based Ergonomic Risk Analysis

Over the last few decades, a number of postural profiling and evaluation systems, such as the Ovako Working Posture Assessment System (OWAS) (Karhu et al. 1977); the Rapid Upper Limb Assessment (RULA) (McAtamney and Nigel Corlett 1993); the Rapid Entire Body Assessment (REBA) (Hignett and McAtamney 2000); and the Posture, Activity, Tools, and Handling (PATH) method (Buchholz et al. 1996) have been introduced in the literature; however, practical deployments and evaluations using these systems are often conducted manually. Ray and Teizer (2012) present one of the earliest methods for automatic worker ergonomic analysis based on visual data. Their method uses red–green–blue–depth (RGBD) sensors and estimates various body postures of a worker such as raised arms, standing, squatting, bending, and crawling by checking the 2D configurations of predicted worker body keypoint locations. However, the estimated poses are directly used to recognize safety incidents without measuring their severity. More recent works on visual ergonomic assessment feed visually estimated poses and joint angles to the OWAS, REBA, and RULA evaluation systems. For instance, Massiris-Fernández et al. (2020) consider the single-camera positions to adjust estimated 2D keypoint locations and then estimate RULA scores. Liu et al. (2016) propose a 3D human skeleton extraction method from a stereo video camera. Some of the recent works have reconstructed a set of 3D keypoint locations based on the image-based recognition of the body posture. For instance, Zhang et al. (2018), Yu et al. (2019), and Chu et al. (2020) estimate 2D keypoint locations first and transform the 2D keypoints to 3D coordinates using various pretrained 2D to 3D human pose conversion models. The resulting 3D worker skeletons are used to generate OWAS pose codes or to calculate REBA scores. High interobserver agreement is achieved between the visual-based REBA scores and REBA scores generated from wearable inertial measurement units (IMU). Their work shows that existing

visual-based pose estimation models can be reliably used for localizing workers' 2D and 3D body keypoints and generating pose assessment scores. However, one common limitation of the aforementioned work is that the context of worker activity and workplace is not considered as a part of the ergonomic analysis. Similar to the PATH method—which extends the OWAS approach by also considering workers' activity and tool use—this paper leverages worker pose, activity, PPE, tools, material, and context, simultaneously; however, different from PATH, this process is entirely automatic. As a simple illustration of this idea, the OWAS system is used and the pose classification of Zhang et al. (2018) is followed. As such, this paper contributes to the body of knowledge in visual-based ergonomic assessment by extending these methods and specifically unifying the recognition with PPE compliance risk analysis as well as activity recognition.

Computer Vision-Based Worker Activity Recognition

Construction resource activity recognition is a well-studied topic (Azar and McCabe 2012; Memarzadeh et al. 2012; Cheng and Teizer 2013; Han et al. 2013; Golparvar-Fard et al. 2013; Khosrowpour et al. 2014), and a more comprehensive review can be found in Yang et al. (2015). Recent computer vision methods recognize worker activities from video feeds by applying conventional computer vision model architectures and pipelines that rely on hand-selected features. Yang et al. (2016) use dense trajectory features to classify worker activities in recorded video clips. Luo et al. (2018b) apply temporal and spatial streams to classify workers' activity in recorded construction site videos. Luo et al. (2018a) apply RGB image, optical flow, and grey image streams to classify worker activities in a steel reinforcement task. Ding et al. (2018) integrate a 2D convolutional neural network and a long short-term memory network to recognize unsafe actions in recorded worker climbing ladder videos. Luo et al. (2019a) apply a 3D convolution neural network on tracked workers and label activity for each worker tracklet. Luo et al. (2019b) use a probabilistic graphical model to refine worker activity labels in untrimmed videos. Other more recent works explore worker activity recognition assisted by other high-level visual information. Cai et al. (2019) improve group activity recognition by using worker head orientations as attention cues. Roberts et al. (2020) simultaneously perform activity recognition and worker pose tracking in untrimmed videos and show that activity recognition can be improved by incorporating recognized poses. This paper builds on the same idea of Roberts et al. (2020) to jointly perform detection and activity recognition, but it is different in many ways. First, object features for tools are added along with pose features to assist activity recognition. Second, the present study also performs frame-wise activity recognition but uses single images instead of stacks of images as input. This allows a unified visual model to simultaneously generate object, activity, and pose predictions; hence, much less computation resource is needed. Luo et al. (2019b) are close to the present study in terms of probabilistic graphical modeling of activity recognition, but their method does not use pose, tool, equipment, or material information as the basis of their recognition.

The present study applies a spatio-temporal graph modeling for activity recognition that is inspired by the structural recurrent neural network (RNN) (Jain et al. 2016). The introduced structural RNN recognizes human activities by modeling humans and objects as nodes in a graph, pairwise human-object and object-object relations at a time step t as spatio-temporal edges and transition of the same node through time as temporal edges. With the goal of predicting nodes' labels (i.e., human activity), this architecture uses a factor graph formulation to model each node's factor

function and each pairwise edge's factor function. The graphical formulation of context proposed in this study is tailored to treat worker activity, pose, and interactions with tools and materials at time step t as nodes and nodes' relations as edges. In the "Results and Analysis" Section, it is shown that this new formulation significantly improves frame-wise activity recognition.

Method

Risk Analysis Model Using Multi-Type Visual Information

A new single-worker risk analysis model is proposed in this study, and the workflow is illustrated in Fig. 1. The input of the workflow is a stack of consecutive frames of a single-worker video taken at a close distance. In step 1, the model first processes every image from the stack and produces initial predictions of worker activity, PPE, tools, material, and 2D body keypoints. In step 2, a spatio-temporal model refines activity recognition from per-frame recognition results. In step 3, the estimated 2D body keypoints are lifted to 3D coordinates using a pretrained keypoint conversion model (Pavlo et al. 2019). In step 4, each single frame's safety severity level per worker is predicted with a trained classifier taking the visual recognized results. With these four steps, the worker risk score can be calculated. In this paper, the proposed risk analysis model is validated with a single-worker outdoor bricklaying and plastering image dataset. To extend to other single-worker construction activities, the workflow requires new images taken with similar worker-camera configurations to images used in this study as well as image annotations using the presented format. Models trained in this study can be fine-tuned to other activities to reduce data size requirements.

Risk Analysis Formulation

The risk formulation in Årnes et al. (2005) is followed by using a discrete state space and a function that maps a state to a severity level. The state space describes a single-worker's *state* at time t . A worker state is defined as a vector with nine components: (1) three components from the pose codes of worker arms, back, and legs; (2) one component containing worker activity label; (3) two components for detecting hard hat and gloves as two required worker PPE; (4) two components for tools and material interactions; and (5) one component for work context to indicate potential falling to a lower level hazard.

$P(state)$ denotes the probability distribution of worker states when a worker's body pose, and the presence of PPE, tools, and materials are observed in a visual frame. $S(state)$ is the severity

level of workers in a state. An assumption is made that components are independent given the input image. This allows a simplification in modeling state probability byproducts of individual probability instead of a joint probability space. Then, $P(state)$ can be evaluated using visual recognition model predictions such that $P(state) = \prod_i P(i)$, and i is a component in the worker state. Thus, risk score (R^t) is computed through Eq. (1)

$$R^t = E[P(state)S(state)] = \sum_j S(j) \prod_i P(i) \quad (1)$$

where j means a possible worker state generated by visual models' predictions. Because of the high dimensionality in the worker state space and the high confidence outputs of deep learning models, in practice, the distribution $P(state)$ is peaked at one state. An example of this phenomenon is shown in Fig. 2. Since the probability mass is concentrated in a few possible worker states, Eq. (1) can be approximated by considering a few top likely worker states, i.e., those who allocate more than 95% probability mass. Computation of a worker state probability is explained in the "Experiment and Implementation Details" section.

Worker Postural Code Using the OWAS System

In this method, a worker's arm, back, and leg poses are classified similar to the OWAS pose profiling (Karhu et al. 1977) and its later implementation for visual-based ergonomic assessment (Zhang et al. 2018; Yan et al. 2018). Fig. 3 shows the worker's body keypoints arrangement and pose classification angles. Given the estimated gravity direction, poses are classified based on the angles between skeleton (vectors that connect keypoints) and the gravity direction [see Back1, Back2, Arms, Legs angles in Fig. 3(b)]. Pose code criteria are listed in Table 2, following recommendations in Zhang et al. (2018). Fig. 4 shows two examples of arms poses. More details on the estimated gravity direction are presented in the Experiment and Implementation Details Section.

Machine Learning-Based Severity Level Classification

The worker-level severity $S(state)$ of a worker state is estimated with a classification model trained on ground truth human ratings. Participants rate one of the four severity levels: *Negligible*, *Low*, *Medium*, and *High*. These severity levels (Table 3) are defined similarly to OWAS's action categories. A logistic regression model is fitted using worker state features and human severity level ratings. These severity levels can be converted to a normalized severity weight from 1 to 100 for calculating the final risk score, similar to the method of Rozenfeld et al. (2010). Severity weights should be adjusted by safety personnel when the method is applied to other

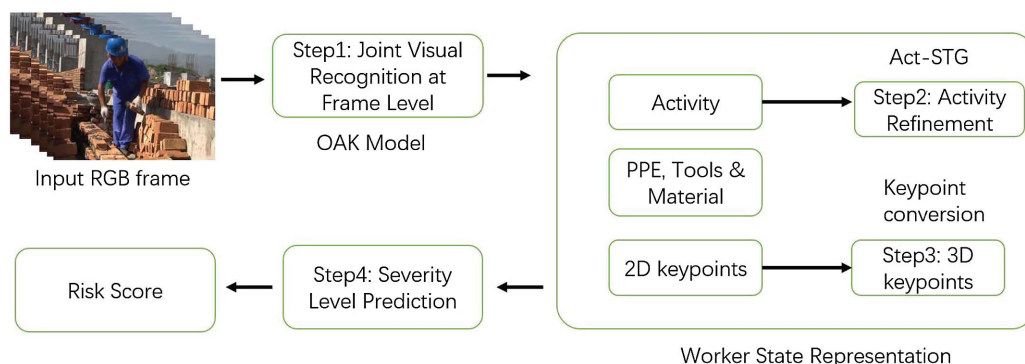


Fig. 1. Workflow of the presented risk analysis model. (Image data from Yang et al. 2016; image courtesy of Jun Yang.)

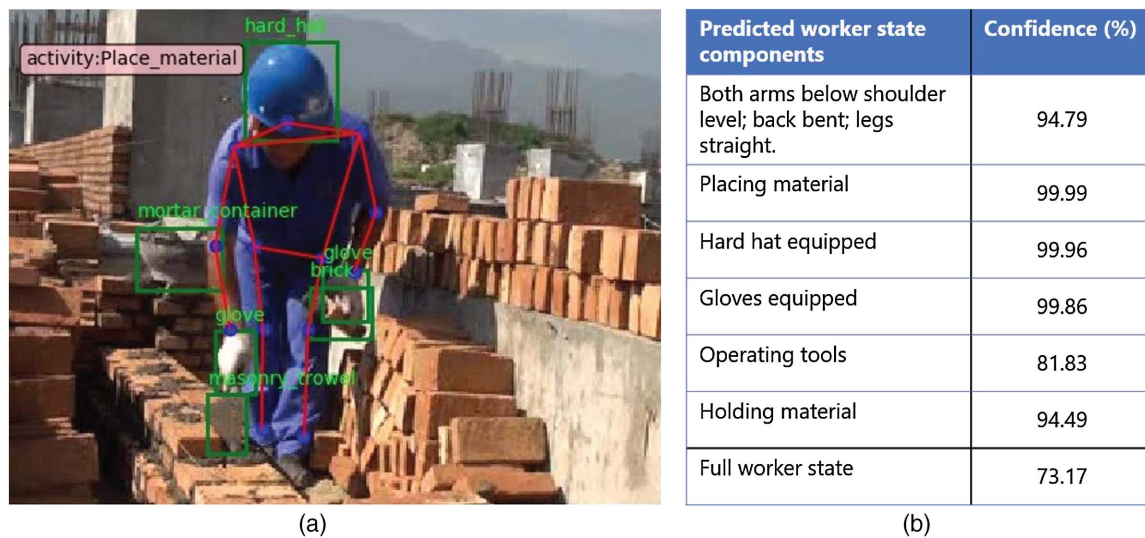


Fig. 2. An example of the highest confidence worker state predicted by visual models: (a) visualization of model predictions on objects, activity, and 2D body keypoints that define the mostly likely worker state; and (b) prediction confidence scores of the mostly likely worker state's components. (Image data from Yang et al. 2016; image courtesy of Jun Yang.)

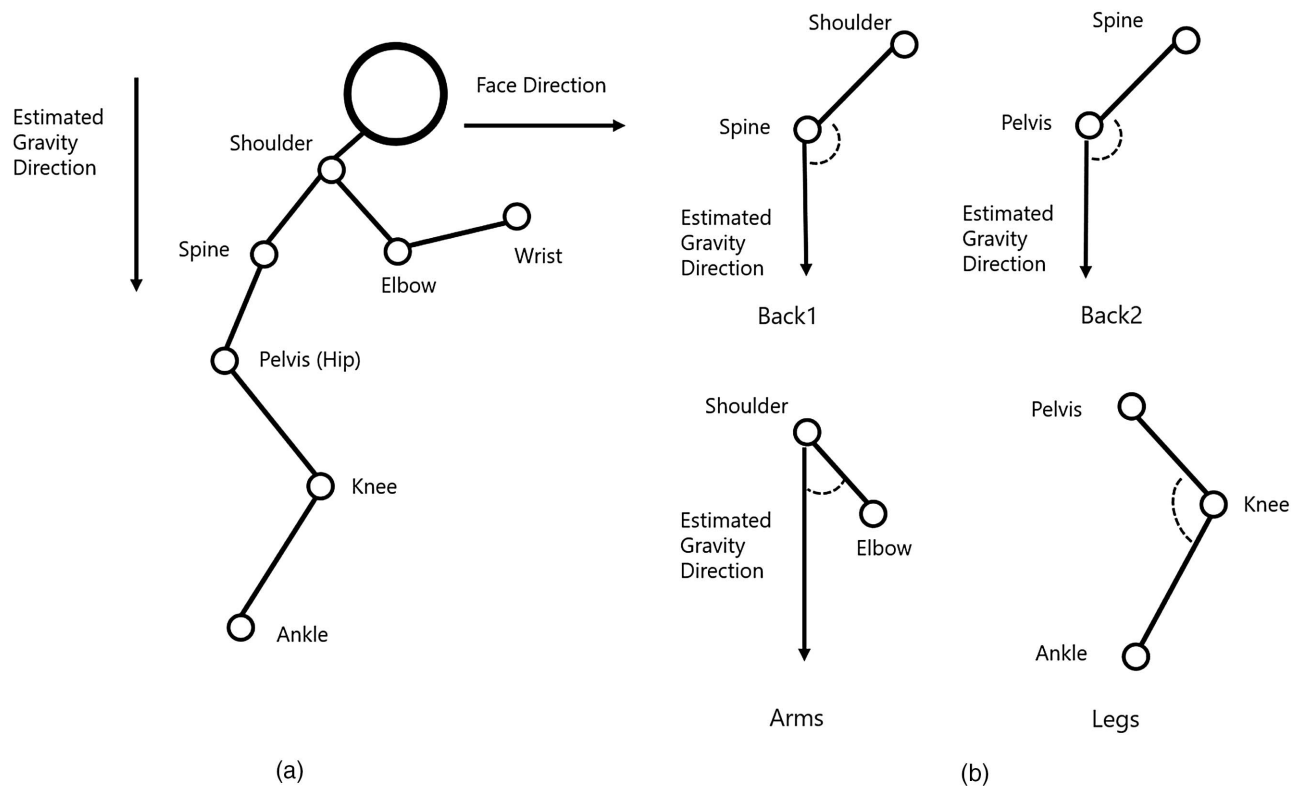


Fig. 3. Worker body keypoints used for worker pose classification: (a) body keypoints arrangement; and (b) shows four body joint angles used to classify a worker's pose: the top two subfigures are for a worker's back; lower left subfigure is for a worker's arms; lower right subfigure is for a worker's legs.

projects. More details in obtaining ground truth severity level annotation are shown in the Experiment Setup and Dataset Section.

Visual Recognition Models

Two types of models are developed to automatically recognize workers' activities, poses, PPE use, and tool and material interactions from video frames. The first model is a modified version of

Mask RCNN (He et al. 2017), which in this paper it is referred to as the Object-Activity-Keypoint (OAK) RCNN model. This model performs the following tasks simultaneously: detecting worker, material, PPE, and tools; recognizing per-frame worker activity; estimating a worker's 2D body keypoints' locations. The second is a spatio-temporal graph neural network model that refines per-frame activity recognition results by connecting recognition results across frames. This model is referred to as Act-STG.

Table 2. Pose classification using the four types of angles

Body part	Positions	Angle range
Arms	A: Both arms on or below shoulder level B: One arm on or below shoulder level C: Both arms above shoulder level	Both Arms angles $\leq 90^\circ$ One Arms angles $> 90^\circ$ Both Arms angles $> 90^\circ$
Back	A: Back straight B: Back bent C: Back bent heavily	Both Back1 and Back2 angles $\geq 160^\circ$ Back1 or Back2 angle $\in [120^\circ, 160^\circ)$ Back1 or Back2 angle $\in [0^\circ, 120^\circ)$
Legs	A: Standing straight B: Knee bent C: Squatting	Both Legs angles $\geq 160^\circ$ At least one Legs angle $\in [60^\circ, 160^\circ)$ Both Legs angles $< 60^\circ$



(a)



(b)

Fig. 4. Snapshots of arms positions during plastering activity: (a) applying plaster with both arms on or below shoulder level; and (b) applying plaster with one arm above shoulder level. (Image data from Yang et al. 2016; images courtesy of Jun Yang.)**Table 3.** Severity level definitions. Severity weights are user-defined normalized values from 1 to 100

Severity level	Severity weight	Definition
Negligible	1	Activity and workplace context do not cause harm to the worker's head, arms, hands, back, or legs. No action required.
Low	5	Activity and workplace context may cause slight but no apparent harm to the worker's head, arms, hands, back, or legs. The worker must be checked in the near future.
Medium	25	One of the head, arms, hands, back, and legs is subject to acute or chronic harm caused by worker's activity and workplace context. Corrective actions are required as soon as possible.
High	100	Two or more of the head, arms, hands, back, and legs are subject to acute or chronic harm caused by worker's activity and workplace context. Immediate corrections are required.

Object-Activity-Keypoint RCNN Model

The model's network architecture for per-image object detection, activity classification, and keypoint estimation, is shown in Fig. 5. The OAK model is based on a generalized RCNN framework and consists of four parts: the backbone feature maps, an object

detector, a keypoint estimator, and an activity predictor. For backbone feature maps, a pretrained Resnet50 + FPN (Lin et al. 2017) convolutional neural network is used as model initial weights; the output image feature (*layer4* feature) is a 3D tensor whose depth is 256 and height and width are proportional to image height and width. The object detector is a Fast RCNN model with a region of interest pooling (RoIAlign) layer (He et al. 2017); the pooled regional features are used to predict object bounding boxes' locations and classes [Fig. 5(a)]. The activity recognition head pools the regional features from the whole convolutional feature maps [Fig. 5(c)]. The keypoint estimator is a variation of Mask RCNN with a keypoint estimation head [Fig. 5(b)]. The keypoint estimation head takes a detected worker bounding box and pools its regional features again. These regional features are fed to a mininet-work consisting of four convolutional layers and one transpose convolutional layer. The keypoint head output is a 4D tensor of size $1 \times C \times W \times H$, with C as the number of keypoint classes and W and H being the regional feature maps' height and width; in the present study, they are both 56. For each channel in C , the location (w, h) with the highest confidence indicates the keypoint's location in the feature map. A final transformation is used to convert keypoint locations from feature map coordinates to image coordinates. The 3D keypoints' locations are generated using an off-the-shelf keypoint conversion model (Pavlo et al. 2019) that can take 13 2D body keypoints to 17 3D body keypoints. Note that

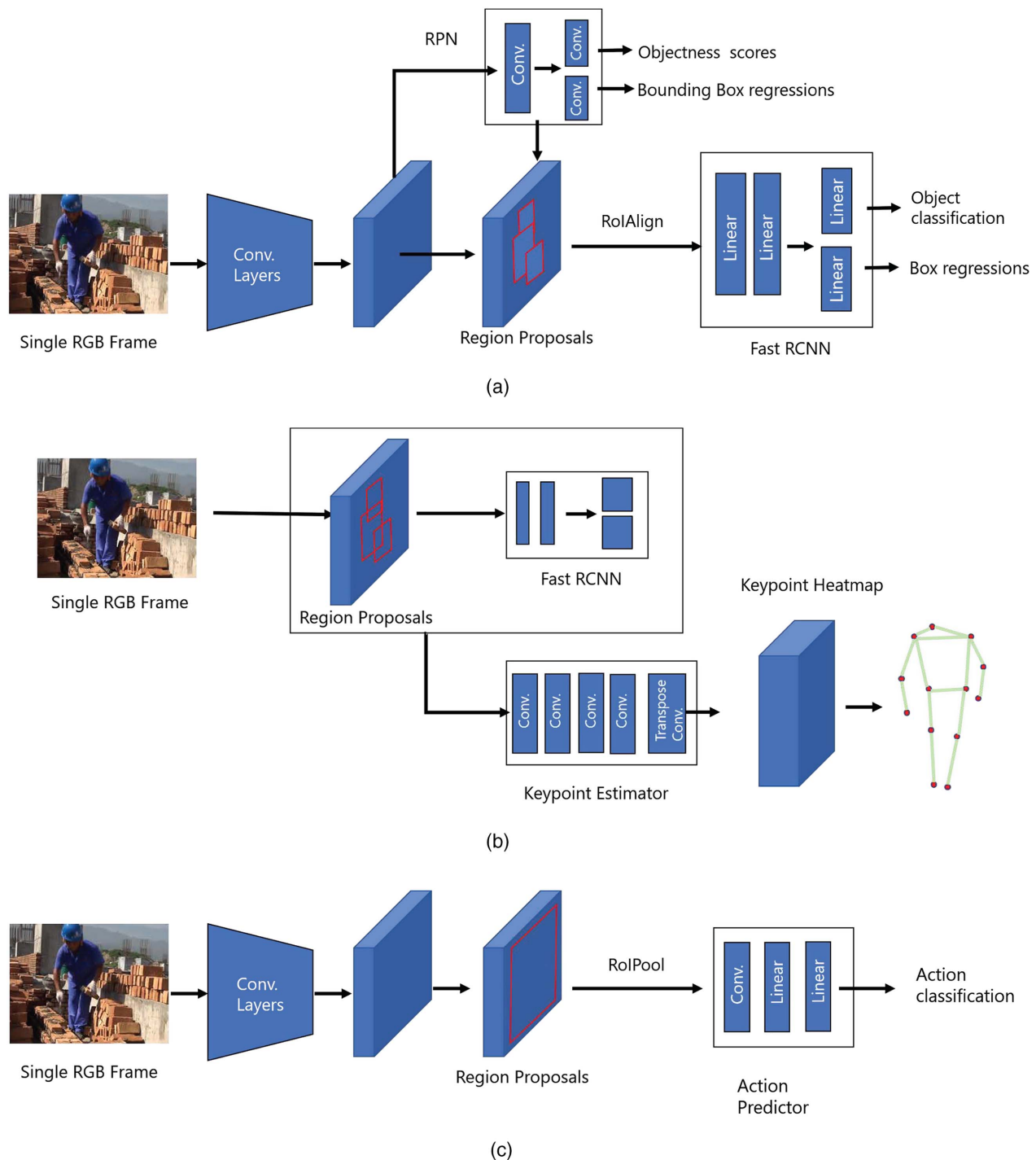


Fig. 5. OAK model architecture. All modules share the same input image and backbone convolutional feature map; each task is performed by a specific head: (a) object detection module; (b) keypoint estimation module; and (c) activity classification module. (Image data from Yang et al. 2016; images courtesy of Jun Yang.)

the three heads reuse the backbone convolutional features in both training and testing phases. This treatment significantly reduces model size and inference time compared to training three separate models while keeping a comparative performance. More details are presented in the “Results and Analysis” section.

Spatio-Temporal Graph Model for Activity Recognition (Act-STG)

A spatio-temporal graph $G = (V, E_S, E_T)$ is composed of a collection of random variables V that are associated with nodes, a

collection of spatio-temporal edges E_S between a pair of nodes at time t , and a collection of temporal edges E_T connecting the same node from time t to time $t + 1$. Such a graph models a probability distribution of nodes and is used to predict node labels or real value vectors y at time t (Jain et al. 2016). This formulation is considered in the present study to refine activity recognition (Fig. 6) with per-frame activity, worker body keypoints, and detected tools and materials as nodes in the graph. Jain et al. (2016) apply a factorization of this graph to simplify marginal probability computation with a factor function $\Psi(y_v, x_v)$ for each node and a factor

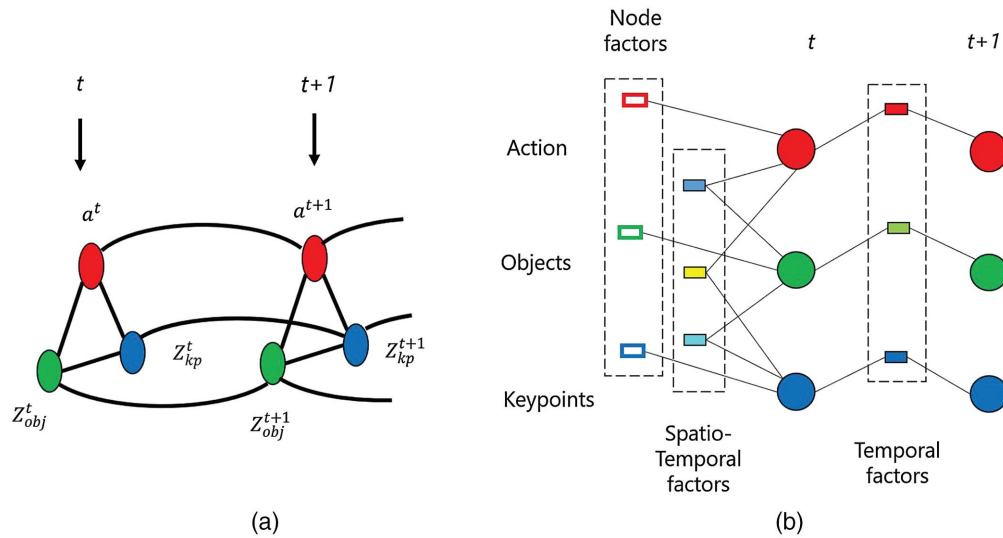


Fig. 6. Structure RNN model for spatio-temporal modeling: (a) proposed spatio-temporal graph; and (b) a factor graph representation of the proposed spatio-temporal graph.

function $\Psi(y_e, x_e)$ for each edge; x_v and x_e are feature representations for each node and edge, respectively.

As shown in Fig. 6(a), to construct a model that has a scalable representation of an arbitrary size graph, a pseudo node is introduced representing the summarized features for the detected tools and materials [Eq. (2)], where f_{MLP} is a single feedforward network to map bounding box regional features to graph node features, and Z_{obj} is the final object feature averaged from all objects. A similar pseudo node is introduced for all estimated worker body 2D keypoints [Eq. (3)] to generate a summarized keypoint feature Z_{kp} for workers at each frame. Vector a in Fig. 6(a) is the pooled regional activity feature from the OAK model. When Fig. 6 is viewed in color, the red color refers to activity nodes and factors, the green color refers to object nodes and factors, and the blue color refers to keypoint nodes and factors. Note that to reduce overfitting, predicted activity scores from the OAK model are not used in Act-STG as a feature

$$Z_{obj} = \frac{\sum n_i}{N}; \quad n_i = f_{MLP}(obj_i) \quad (2)$$

$$Z_{kp} = \frac{\sum k_i}{K}; \quad k_i = f_{MLP}(kp_i) \quad (3)$$

Similar to structural RNN (Jain et al. 2016), a factor graph [Fig. 6(b)] is used to formulate the spatio-temporal graph [Fig. 6(a)]. Nine factors are defined in total: three node factors for activity (f_{act}), objects (f_{obj}), and keypoints (f_{kp}); three spatial edge factors connecting activity and objects (r_{ao}), activity and keypoints (r_{ak}), and objects and keypoints (r_{ok}); and finally three temporal edge factors that model time propagation of activity (r_a), objects (r_o), and keypoints (r_k). While the spatio-temporal edge factors capture the relations of recognized activity, detected objects, and estimated keypoint locations between frames, the node factors refine per-frame activity using the spatio-temporal features. All factors in the present study are implemented with a gated recurrent unit (GRU) (Chung et al. 2014). Like many graph neural network models, the temporal update rules from time t to time $t+1$ for Act-STG are defined as follows:

$$\begin{aligned} e_{ao}^{t+1} &= r_{ao}(s_a^t + s_o^t) \\ e_{ak}^{t+1} &= r_{ak}(s_a^t + s_k^t) \\ e_{ok}^{t+1} &= r_{ok}(s_o^t + s_k^t) \end{aligned} \quad (4)$$

And the spatial update rules are

$$\begin{aligned} e_k^{t+1} &= r_k(Z_{kp}^{t+1} + e_{ak}^{t+1} + e_{ok}^{t+1}) \\ e_o^{t+1} &= r_o(Z_{obj}^{t+1} + e_{ok}^{t+1} + e_{ao}^{t+1}) \\ e_a^{t+1} &= r_a(a^{t+1} + e_{ao}^{t+1} + e_{ak}^{t+1}) \end{aligned} \quad (5)$$

where e = output of GRU; and s = hidden state of each factor. The final activity prediction F is a summation of all node factor predictions

$$F^{t+1} = f_{act}(e_a^{t+1}) + f_{obj}(e_o^{t+1}) + f_{kp}(e_k^{t+1}) \quad (6)$$

Experimental Setup and Dataset

The image data used in the present study is originated from a collection of video clips first introduced by Yang et al. (2016). The original video resolution is 240p. The majority of the original video collection captured single-worker outdoor construction activities and was taken at a close distance to workers. Yang et al. (2016) has provided video-level activity labels, such as bricklaying and plastering. Roberts et al. (2020) have augmented a subset of bricklaying and plastering videos and have added frame-wise activity labels and single-worker body keypoints. This work extends the annotations in Roberts et al. (2020) by adding per-frame object bounding box annotations of PPE, tools, and material and video-level workplace context annotation. Bounding box annotations of PPE, tools, and materials were assisted by the data team in Alibaba AI Lab, and their qualities were thoroughly and rigorously reviewed by construction experts to correct missing and mislabeled objects.

Dataset Statistics and Characteristics

Images in the presented dataset contain a high interclass and intra-class variance in terms of worker appearances, workplace backgrounds, and camera viewing angles. In the bricklaying set,

Table 4. Activities and objects for bricklaying and plastering

Task	Activity	Description	Objects
Bricklaying	A: Prepare material	Collect mortar or brick, put mortar on brick, or break brick	Hard hat, gloves, masonry trowel, mortar container, and brick
	B: Place material	Place brick or spread mortar on wall	
	C: Consolidate placement	Tap brick into place or remove excess mortar	
Plastering	A: Collect plaster	Collect plaster with hawk or hand board	Hard hat, gloves, masonry trowel, mortar container, and hawk
	B: Transfer plaster	Transfer plaster between hawk or hand board and trowel	
	C: Apply plaster	Deposit and spread plaster on the wall	

18 workers are captured in 179 videos. In the plastering set, 9 workers are captured in 122 videos. Different workers have distinguishable appearance, motion, and surrounding workplace contexts. Each worker is recorded from more than one camera viewing angle and distance, representing highly diverse motion patterns in images. Images used in this study have a realistic representation of real-world bricklaying and plastering data complexity. Building on previous work, the image dataset presented in this study provides abundant annotations for single-worker activity, worker body keypoints, and PPE, tools, material object bounding boxes, and potential workplace hazard labels. The sheer volume and diversity in images and their annotations represent a challenging construction image dataset first of its kind. In total, this image dataset contains 36,969 worker activity labels, 186,464 object bounding boxes, 480,597 worker body keypoints, and 69 video-level potential workplace hazard labels. Table 4 shows the definitions of activity, PPE, tools, and material for bricklaying and plastering sets. There are 13 body keypoints annotated: *nose, shoulders, elbows, wrists, hips, knees, and ankles*. Qualitative annotation examples are shown in Figs. 7 and 8 for bricklaying and plastering, respectively. Workplace hazard labels indicate the potential of workers falling to a lower level. A detailed annotation breakdown by training and testing sets is presented in the following section.

The majority of images in this dataset focus on outdoor construction activities in commercial projects. However, there are two indoor construction workers in the plastering set, e.g., the right worker in Fig. 8. The camera viewpoint constraint due to small and clutter indoor space can be critical to apply the proposed method on indoor construction images because annotating and training on

OAKs and severity levels are more challenging when a worker's frontal and side views are not available.

Data Split

To validate vision models and the proposed risk analysis method, the dataset is split by workers. For each worker conducting that task, (1) when the number of videos for this worker is even, training and testing videos are randomly split by half; and (2) when the number of videos is odd, extra videos are placed into the training set. Finally, the bricklaying task has 101 videos in the training set and 78 videos in the testing set. The plastering task has 67 videos in the training set and 55 videos in the testing set. The dataset is split in such a way to avoid almost identical images in the training set from “bleeding” into the testing set while diversifying testing data. Vision model validations and risk analysis human calibrations are conducted on the testing set videos. Table 5 shows a detailed annotation breakdown for the training and testing sets.

Worker Severity Level Annotation

Only testing images of bricklaying and plastering sets are used to obtain ground truth annotations of worker severity levels. Since the labeling severity level requires substantial safety knowledge and can be subjective to annotators' own experience, a two-stage annotation is applied to obtain consistent severity level labels. In the first stage, a number of safety experts are invited for an online survey to rate severity levels on 10 randomly selected images. Instruction and severity level definitions are provided in the online survey.

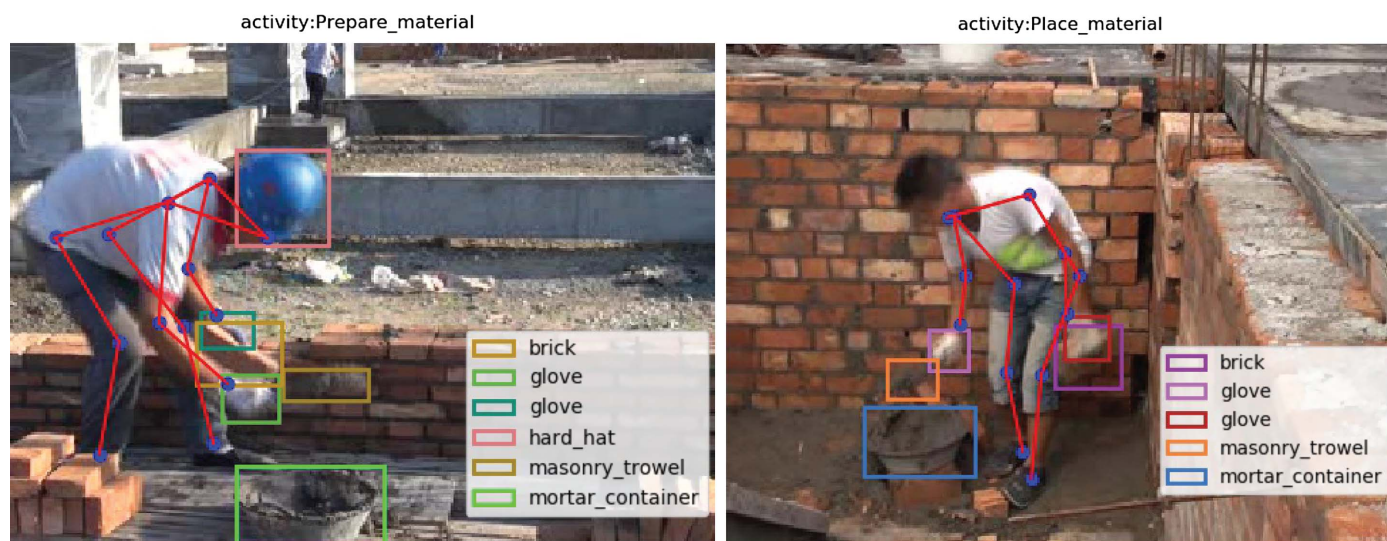


Fig. 7. Annotation examples of bricklaying task. Each frame is annotated with an activity label, a set of body keypoints connected by line segments, and a set of object annotations on PPE and tools. (Image data from Yang et al. 2016; images courtesy of Jun Yang.)



Fig. 8. Annotation examples of plastering task. Each frame is annotated with an activity label, a set of body keypoints connected by line segments, and a set of object annotations on PPE and tools. (Image data from Yang et al. 2016; images courtesy of Jun Yang.)

Table 5. Annotation numbers in training and testing sets

Tasks	Split	Frame-level activity labels	Frame-level object labels	Frame-level keypoint labels	Video-level hazards labels
Bricklaying	Train	12,146	68,370	157,898	32
	Test	9,327	53,806	121,251	20
Plastering	Train	8,829	37,303	114,777	12
	Test	6,667	26,985	86,671	5



Fig. 9. The interface used for the second stage severity level annotation. (Image data from Yang et al. 2016; image courtesy of Jun Yang.)

All annotators are anonymous and encouraged to disclose their years of experience in construction safety. A multiparticipant inter-observer agreement test (Fleiss and Cohen 1973) is conducted to find annotators in close agreement. These annotators are later invited for the second stage annotation where a randomly selected image is presented in an interface (Fig. 9). Annotators are asked to select a severity level for the depicted worker and click the “Next Image” button to save their annotations and proceed to the next image. Each image will be annotated at most once. Annotation ends when all test images are annotated or a participant closes the

interface. All severity level annotations collected in the second stage are used as final ground truth severity labels. In the first stage annotation, 7 participants achieved a 0.64 kappa score. This is close to the 0.7 kappa score of safety index prediction from Poh et al. (2018). There were 4 participants with over 5 years of experience in construction safety and 3 participants with 1 to 3 years of experience in construction safety. In the second stage, the selected participants annotated 189 bricklaying images and 217 plastering images (see severity level examples in Fig. 10). Ground truth severity level annotations statistics are shown in Table 6.



Fig. 10. Examples of severity level annotations: (a) negligible; (b) low; (c) medium; and (d) high. (Image data from Yang et al. 2016; images courtesy of Jun Yang.)

Experiment and Implementation Details

Experiment Procedure

For each task, an OAK model is first trained, then an Act-STG model, and then the trained models are applied for the testing set. Predicted 2D keypoint locations are fed to the 2D-3D keypoint conversion model to generate 3D keypoint locations. Finally, predicted activities from Act-STG, detected PPE, tools, and material from OAK, estimated 3D keypoints, and ground truth work context conditions are combined as worker states. A subset of testing images for both tasks is evaluated by humans, and estimated worker states for these images are used to fit logistic regression models.

Table 6. Numbers of ground truth worker severity level annotations

Task	Negligible severity	Low severity	Medium severity	High severity
Bricklaying	33	88	58	10
Plastering	45	42	96	34

All training and testing are run using a system with an Intel i7 8700K processor, an Nvidia RTX 2080Ti graphics card, and 32 Gb memory.

Evaluation Metrics

Due to being computationally intensive, the inference speed, memory consumption, and the performance of the OAK model are evaluated first. Per-frame activity prediction of Act-STG and severity level classification are evaluated later as a high-performing visual recognition model is essential for the severity classification. The model performance metrics are defined as in the following.

A correctly recognized instance is called a true positive (TP). A falsely recognized instance is called a false positive (FP). A missed ground truth instance is called a false negative (FN). Precision is defined as the ratio between TP and TP + FP, and recall is defined as the ratio between TP and TP + FN. By adjusting the threshold for prediction confidence, a precision-recall curve of a class is generated. The average precision (AP) is defined as the area under the precision-recall curve, mean AP (mAP) is defined as the average of AP for all classes. The mAP metric evaluates the overall model

performance across all possible prediction score thresholds; it is a robust metric focusing on the general model performance (Lin et al. 2014; Ren et al. 2015; He et al. 2017).

For activity recognition, mAP is reported using activity scores of all classes and the ground truth activity class label each frame. Precision and recall averaged over activity classes are reported, and in this case, a frame-wise activity prediction is the class having the highest activity score. The mAP for object detection and keypoint estimation should be reported considering localization errors. The intersection over union (IoU) is defined as the ratio between the intersection area and the union area between a detected bounding box and a ground truth bounding box. In the present study, a 0.5 IoU threshold is used to determine whether a ground truth object instance is matched to a detected object instance. Thus, object detection is evaluated by mAP at IoU = 0.5. For keypoint estimation, the object keypoint similarity (OKS) (Lin et al. 2014) is defined in Eq. (7)

$$OKS = \frac{1}{N} \sum_i \exp\left(\frac{-d_i^2}{2s^2k_i^2}\right) \quad (7)$$

where i = i th keypoint; N = total number of keypoints; d_i = Euclidean distance between a ground truth keypoint and its corresponding detected keypoint; s = person bounding box area in pixels; and k_i = scaling factors for each type of keypoint: 0.026, 0.079, 0.072, 0.062, 0.107, 0.087, and 0.089 for the nose, shoulders, elbows, wrists, hips, knees, and ankles, respectively (Lin et al. 2014). In the present study, a worker's pose is correctly detected if its OAK is greater or equal to 0.5; hence, the pose evaluation metric is mAP at OAK = 0.5.

Regressing severity level is evaluated by the 5-fold cross-validation accuracy. This accuracy is computed by averaging the accuracy of each fold while treating the remaining four folds as training data. The standard error on cross-validation accuracy is used as a measurement of deviation.

Computing Worker State Probability

Worker state probability is computed by the product of components probabilities. For example, the probability of hard hat use is the detected hard hat box's probability p_h if such box's exterior is within 50 pixels of the nose's 2D location; otherwise, it is $(1 - p_h)$. The same 50-pixel criterion is applied for gloves, tools, and materials. This criterion is a hyperparameter that can be selected by users. Using 50 pixels works well in the presented dataset as all images only contain a single worker and have dimensions of 320-by-240 pixels. For crowded multiple worker images, a learned interaction recognition model is recommended (Tang et al. 2020). The probability of work context is zero or one because it is treated as given information apart from the visual model outputs. The probabilities of arms, back, and legs are computed based on their corresponding keypoints' prediction scores. For example, the probability of arms' pose $P(arms)$ is the product of the left and right elbows' probability and the left and right shoulders' probability. An estimated keypoint's probability is computed as the summed probability over a rectangle box centered at the estimated location because any keypoint estimation within a certain distance of its ground truth location is not penalized in the OKS keypoint evaluation. The size of the rectangle box is computed by using 10% of the worker box's height and width.

Implementation Details

As previously mentioned, the OAK model is modified based on a Pytorch vision package implementation of Mask RCNN. The joint training of object detection and keypoint estimation is achieved by

masking out keypoint targets for nonworker objects. The training process is improved by initiating the backbone feature weights with a pretrained Microsoft Common Object in Context (MS-COCO) (Lin et al. 2014) person keypoint estimator. For both tasks, OAK models are trained with stochastic gradient descent (SGD) optimizer with 0.9 momentum. The learning rate is set as 0.001 with 0.1 decay at every five epochs, and weight decay is set as 0.0005 for each epoch. There are nine epochs trained to obtain the best models, and the Act-STG model is also implemented using Pytorch. For each task, the Act-STG model is also trained with an SGD optimizer and 0.9 momentum. The learning rate is set as 0.00001. Gradient clipping is set as 50% of the gradient norm to achieve a stable training process. An off-the-shelf 2D-3D person keypoint conversion model from the official code repository of Pavllo et al. (2019) is used; the gravity direction is assumed to be $v = (0, 0, -1)$. Two qualitative examples of keypoint conversion are shown in Fig. 11.

Results and Analysis

OAK Model Analysis

First, all models are profiled using the official Pytorch vision metric logger tool. At inference time, the OAK model runs at 26.6 frames per second and takes 887 MB memory, making our method real-time in practice. Compared to a system running individual models for each subtask (Bundled Model in Table 7), the OAK model takes 36.6% less time and 40.1% less memory. The OAK model inference speed may reach 50 to 100 frames per second if further engineered with acceleration toolkits such as TensorRT. Act-STG and Keypoint conversion models add negligible inference overhead. These results show that the proposed workflow has the potential to process images at near real-time speed.

OAK model's improvements in inference speed and space usage are at small or even no cost of model performance. Table 8 shows that the OAK model achieves 83.2% mAP at 0.5 OKS for keypoint estimation on the bricklaying task, losing 0.5% mAP to the keypoint RCNN model. The OAK model achieves 88.8% mAP at 0.5 OKS for the plastering task, outperforming the keypoint RCNN model by 1.6% mAP. Even though the OAK model adds one parallel task for activity and detects multiple nonworker object classes, it still achieves comparable keypoint estimation capability to the original keypoint RCNN on this construction worker dataset. Table 9 shows a similar trend for object detection performances. The OAK model's object detection performance is very close to that of the faster RCNN model. The OAK model achieves 78.7% mAP at 0.5 IoU for the bricklaying task and 69.6% mAP for the plastering task, while the faster RCNN model achieves 78.1% mAP and 70.6% mAP, respectively. Surprisingly, the OAK model gains small improvements in activity recognition over the activity model. In Table 10, the OAK model reaches 62.5% mAP for the bricklaying task, outperforming the activity model by 0.7% mAP, and it also reaches 75.3% mAP for the plastering task, outperforming the activity model by 0.5% mAP. Precision and recall results between the activity model and the OAK model are also comparable, although the OAK model shows a slight advantage in precision or recall. These results show that the proposed strategy of merging low-level image features from different vision tasks into a common set of convolutional feature maps does not significantly reduce model performance. In other words, the three branches in the OAK model share a common set of convolutional backbone features, so the low-level image features are informed by three different tasks. However, the high-level task features in their individual classifiers

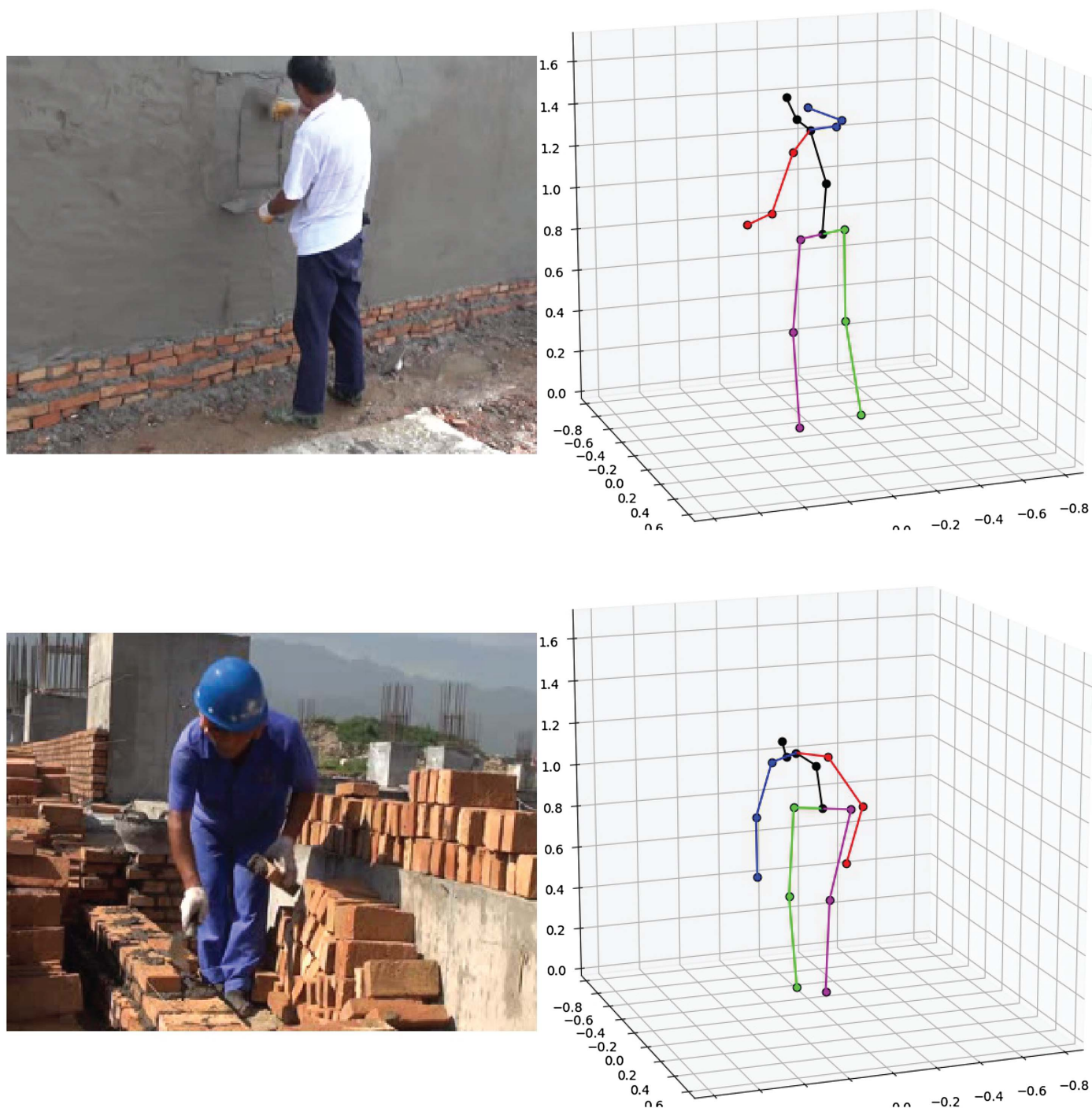


Fig. 11. Qualitative examples of 2D keypoint conversion to 3D keypoints. Examples show how the reconstructed 3D body joint keypoints accurately reflect workers' poses. Values of the reconstructed 3D keypoint locations are proportional to the actual locations'. (Image data from Yang et al. 2016; images courtesy of Jun Yang.)

Table 7. Model consumed resources

Model	Averaged inference speed (ms)	Model GPU usage (MB)
Activity model	12.3	438
Faster RCNN	22.0	453
Keypoint RCNN	25.0	590
Bundled models	59.3	1,481
OAK model	37.6	887

Table 8. Keypoint estimation evaluation results

Model	Keypoint mAP (oks = 0.5) (%)	
	Bricklaying	
Keypoint RCNN		83.7
OAK model		83.2
	Plastering	
Keypoint RCNN		87.2
OAK model		88.8

are not aware of how other tasks perform, so their communication is indirectly through the low-level convolutional features and backward gradients. This implies two things: the image feature is generalized by three tasks; the convolutional network is over-parameterized such that there is enough space to encode information for three different tasks.

Act-STG Model Analysis

Act-STG model fuses single-frame visual information (i.e., objects, activity, and pose) and refine activity recognition from sequential data. Comparing Act-STG with the best single-frame models in Table 10, the Act-STG model significantly improves activity recognition performance. For the plastering task, 7.6% increment in mAP, 6.2% increment in precision, and 6.9% increment in recall are gained. For the bricklaying task, mAP, precision, and recall performances are raised by 2.3%, 5.5%, and 2.8%, respectively. These

Table 9. Object detection evaluation results

Model	Object mAP (IoU = 0.5)
Bricklaying	
Faster RCNN	77.5
OAK model	78.7
Plastering	
Faster RCNN	70.6
OAK model	69.6

Table 10. Activity recognition evaluation results

Model	mAP (%)	Precision (%)	Recall (%)
Bricklaying			
Activity model	61.8	60.6	56.4
OAK model	62.5	60.6	59.3
Act-RNN	62.6	61.5	60.3
Act-STG	70.1	66.8	66.2
Plastering			
Activity model	74.8	65.7	72.0
OAK model	75.3	67.4	71.7
Act-RNN	76.1	68.7	73.4
Act-STG	77.6	72.9	74.8

Note: Boldface indicates the best results considered for this task.

Table 11. Severity level prediction results

Features	Bricklaying		Plastering	
	Acc. (%)	SE (%)	Acc. (%)	SE (%)
Whole image	18.4	17.1	34.1	13.0
Activity + context	50.5	4.6	54.8	1.3
PPE + tools + material	56.6	2.6	44.2	0.3
Arms + back + legs	60.8	1.0	76.5	1.7
Full worker state	85.7	3.6	86.6	2.6

Note: Boldface indicates the best results considered for that task.

Table 12. The best severity level prediction model performance breakdown by poses and activities

Task	Arms (%)			Back (%)			Legs (%)			Activity (%)		
	A	B	C	A	B	C	A	B	C	A	B	C
Plastering	84.5	86.4	n/a	89.3	82.3	85.0	89.2	78.0	90.4	77.8k	89.6	84.8
Bricklaying	86.4	90.0	n/a	84.6	86.8	87.0	91.4	82.3	n/a	81.4	92.4	85.0

results illustrate the benefits of capturing high-level spatio-temporal relations between different vision information for activity recognition. Act-RNN removes objects and keypoints nodes in Act-STG, it serves as a baseline that simply models temporal activity. Comparing Act-STG and Act-RNN, it is clear that modeling spatio-temporal relations from objects and keypoints contributes to better activity recognition.

Severity Level Prediction Analysis

A generalized linear model framework is used with two constraints: (1) a ridge regression penalty term; and (2) weights are constrained to be nonnegative. Table 11 shows the severity level prediction results, and the best model in the last row uses the full worker state as features. On the bricklaying task, the best model achieves 85.7% cross-validation accuracy (Acc.) and 3.6% standard error (SE). On the plastering task, the best model achieves 86.6% cross-validation accuracy and 2.6% SE. These numbers show that regression models can reliably predict human preference using the generated worker states.

In addition, an ablation study is conducted using variations of worker state components and a fine-tuned ImageNet ResNet18 image classifier that directly predicts the severity level from the whole image. The full comparisons are reported in Table 11. The *whole image* represents the ResNet18 image classifier; *activity + context* refers to the logistic regression model only using activity and context components as the worker state; *PPE + tools + material* refers to the classifier using PPE, tools, and material components as the worker state; *arms + back + legs* refers to the classifier using worker's pose components as the worker state; *Full worker state* is the proposed method. The proposed severity level prediction model using the full worker state achieves the best result. The reported results also show that severity level prediction cannot be easily learned from pure image classification.

The best model for severity level prediction is not significantly biased toward particular worker OWAS body pose codes and worker activities. However, severity level prediction performance is slightly lower when a worker in the state "Knee bent" or "Transfer Plaster." Detailed performance breakdown is presented in Table 12. Note "n/a" under a class in Table 12 means no instance of that class in the images with ground truth severity level annotations.

For every frame in a given video, worker states are recognized and ranked by their probabilities in descending order. The top worker states are counted until their cumulative probability is over 95%. The severity level of each worker state is predicted using the best model and converted to severity weights in Table 3. The proposed risk formulation in Eq. (1) is used to generate a risk score for each frame. Fig. 12 shows two test set videos' risk scores generated from the proposed method. A dot represents the calculated risk score for a frame, worker images are pointed to their corresponding risk scores and frame numbers. These two videos are not used to train severity level prediction models. The worker in the top video is at higher risk in the beginning because of a heavily bent back and not wearing gloves, and later his risk score increases because of applying plaster with one arm raised above shoulder level and

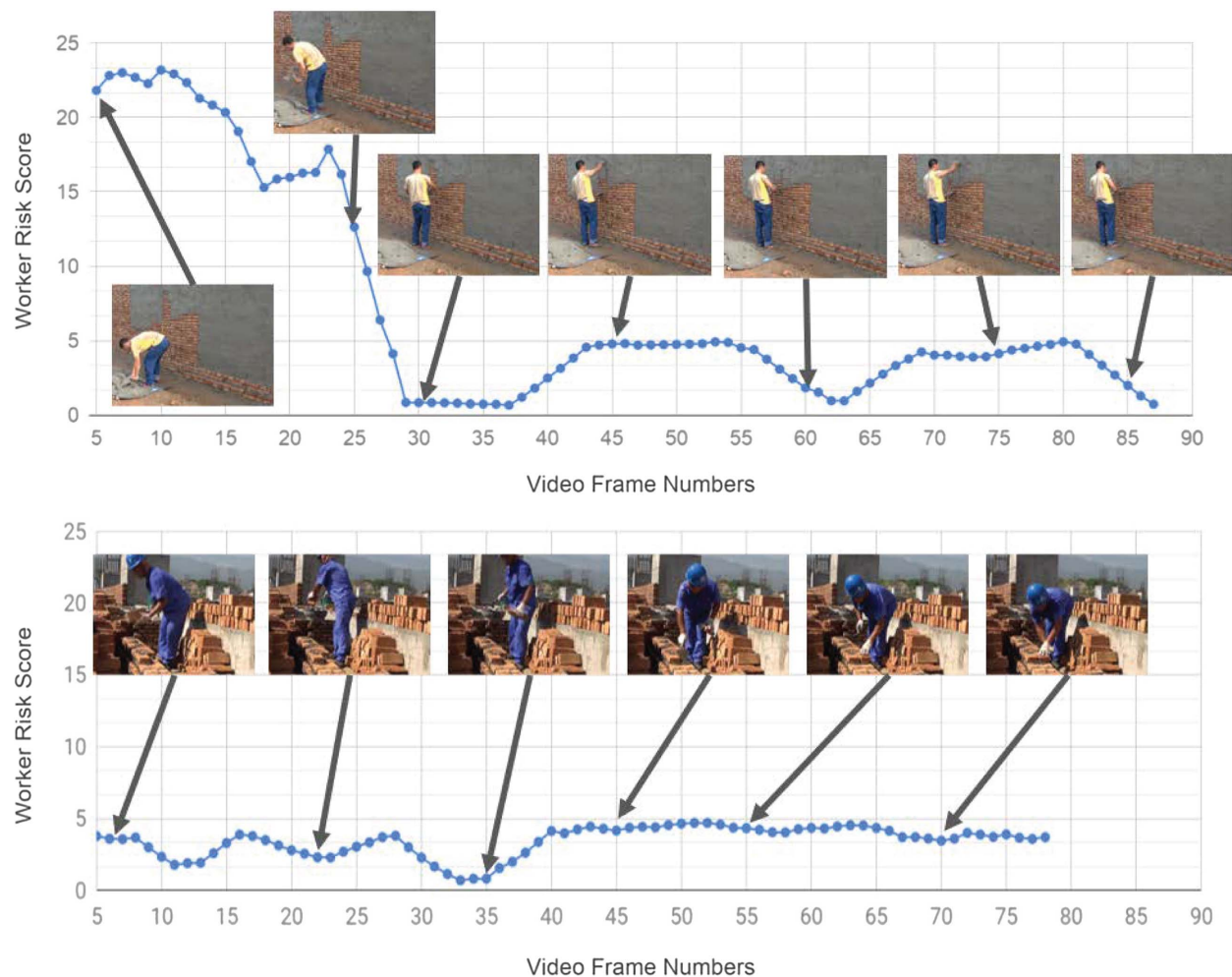


Fig. 12. Calculated risk scores for two test set videos. (Image data from Yang et al. 2016; images courtesy of Jun Yang.)

not wearing gloves. Compare two workers in the top and bottom videos, the bottom worker experiences lower risk because of properly worn PPE, even though his back is bent during placing material.

Limitations and Open Research Areas

The foregoing experiment results and analysis show that automatically obtaining visual knowledge of a worker has a potential to assist worker risk analysis. From a technical standpoint, the present study also shows that abundant visual knowledge of a worker's activity, body pose, PPE use, and interactions with tools and material is beneficial for worker risk analysis and can be efficiently obtained with a multitasked visual model framework.

Nevertheless, there are a few limitations that affect the performance, robustness, and applications of the proposed method.

Data Collection: (1) the proposed method is validated mostly on images that capture single-worker outdoor bricklaying and plastering construction activities at a close distance to workers. To extend this method to other construction site images, such as indoor construction and far-field images, additional image OAK annotations are necessary. However, since the proposed OAK model and Act-STG are derived from generic computer vision tasks, they can be fine-tuned and applied even if only partial subtask data are provided for the new applications. (2) collecting ground truth severity level

labels is still challenging, as substantial safety knowledge and additional interobserver agreement tests are necessary to obtain high-quality labels. Annotators' agreement can shift between companies, regions, and countries. (3) it is needed to avoid wasting annotation resources on the most frequent severity levels because annotator recruitment can be expensive and slow.

Model Development: (1) heavy occlusion of body keypoints and objects can greatly affect severity level prediction outcomes. Robust recognition under heavy occlusion is not considered in this work as the dataset used has low occlusion in general. Robust models to image occlusion are often explored in the context of object tracking, and this is a work for future development. (2) worker pose is a crucial cue for safety professionals to assess risks. In this study, an estimated 3D pose is used for predicting severity level. However, the estimated 3D keypoints are not explicitly evaluated in the present study. We entrust that duty to an off-the-shelf model based on its high performance on generic benchmarks and visual examinations of many construction site image examples. 3D keypoint localization can be better evaluated against IMU-based 3D data (Yu et al. 2019; Chu et al. 2020). Another limitation of the off-the-shelf keypoint conversion model is that camera viewing angles and distances to persons in generic datasets may be different from those in collected construction site images. This difference results in deviations in body pose classification. Future work in this direction will train 3D body keypoint estimation models by construction worker images and their calibrated 3D keypoint locations.

Application: Given the current state of development, the ideal scenario to apply the proposed method is through wearable cameras which are increasingly popular on job sites. This satisfies the requirement of using outdoor, close-ranged worker images. Although the proposed method runs at around 26 frames per second on a single desktop GPU, such a powerful embedded system has not yet been commercially ready. Hence, the proposed method is used offline as a proof-of-concept. Future explorations on system's level topics, such as hardware acceleration and network communication, are needed.

Conclusions

In the present study, a visual-based risk analysis model is proposed for evaluating worker-level risk. The proposed risk model leverages a worker state described by visually obtained worker's activity, worker's body pose, PPE use, worker's interactions with tools and material, and workplace context. A joint learning on object, activity, and keypoints can be efficiently obtained by a unified regional convolutional neural network framework. This strategy significantly reduces the resources required and achieves comparable model performance to models that are dedicated for individual tasks. A spatio-temporal graph neural network model that combines per-frame information on objects, activities, and keypoints is beneficial for improving activity recognition. A machine learning-based severity prediction model takes the worker state and predicts the worker severity level. An ablation study shows that the full worker state is more informative to predict severity level than any worker state using partial visual information. These findings are validated with a newly introduced large image dataset with exhaustive annotations of single-worker activity, body pose, PPE, tools, and materials. The final severity prediction model achieves 85.7% cross-validation accuracy for bricklaying images and 86.6% cross-validation accuracy for plastering images.

Data Availability Statement

Some or all data, models, or code used during the study were provided by a third party: All videos of construction workers used in this study were from Professor Jun Yang (Yang et al. 2016). Direct requests for these materials may be made to the provider as indicated in the "Acknowledgments."

Acknowledgments

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